Undecidability of unnormalized univariate trace inequalities

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Abstract

We prove that no algorithm decides whether a rational polynomial in the ordinary traces of powers of one real symmetric matrix is nonnegative for all finite matrix sizes. The proof encodes nonnegative integers as spectral multiplicities. Stationary polynomial interpolation controls deviations from the encoding nodes with an error bound independent of the number of eigenvalues. A polynomial penalty then transfers an undecidable discrete sign problem to an unrestricted pure trace inequality. The construction uses only positive powers and never uses the matrix dimension as a trace variable.

1 The decision problem

For $d \geq 1$ and $F \in \mathbb{Q}[z_1, \ldots, z_d]$, consider the statement

$$F(\operatorname{tr} A, \operatorname{tr}(A^2), \ldots, \operatorname{tr}(A^d)) \geq 0 \quad \text{for all } n \geq 1 \text{ and real symmetric } A \in \mathbb{R}^{n\times n}. \tag{1}$$

The trace is ordinary and unnormalized. The polynomial is pure trace: its value is a scalar, with no additional matrix factor, and its scalar constant 1 is not $\operatorname{tr} I = n$. By diagonalization, (1) is equivalent to

$$F(p_1(x), \ldots, p_d(x)) \geq 0 \quad \text{for all } n \geq 1 \text{ and } x \in \mathbb{R}^n, \qquad p_r(x) = \sum_{a=1}^{n} x_a^r. \tag{2}$$

Acevedo, Blekherman, Debus and Riener prove undecidability with several matrix variables and decidability for normalized traces [1, Theorems 6.2 and 6.5]. Their Section 7 leaves the ordinary univariate case open. The positivity results of Klep, Pascoe and Volčič use normalized traces [2, Introduction, arXiv version v2]. The distinction is relevant here: ordinary traces retain integer multiplicities, whereas normalization divides those multiplicities by the dimension.

Theorem 1. *There is no algorithm that, on every input consisting of a positive integer d and a polynomial $F \in \mathbb{Q}[z_1, \ldots, z_d]$, always terminates and correctly decides* (1).

We prove the theorem by reducing nonnegativity on integer tuples to (2). A spectrum supported on $\{0, 1, \ldots, k\}$ can record k arbitrary integer multiplicities. The main point is to control every spectrum away from these nodes, uniformly in its size. Interpolating functions with zero first derivatives at the nodes give the required control. Including 0 as a node makes the relevant polynomial sums independent of zero padding and avoids any use of a zeroth power sum.

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2 A uniform estimate for spectral multiplicities

Write $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. Fix $k \geq 1$. For each real x, let $j(x)$ be a closest element of $\{0, \ldots, k\}$, choosing the smaller index at a tie. For a rational polynomial $q(x) = \sum_r b_r x^r$ and $R \geq 1$, use the coefficient bound $\|q\|_R = \sum_r |b_r| R^r$.

Lemma 2. *One can effectively construct rational polynomials $e_1, \ldots, e_k, h$ and a positive rational number C, depending only on k, such that*

$$\deg e_i \leq 2k+1, \qquad \deg h = 2k+2, \qquad e_i(0) = h(0) = 0, \qquad h(x) \geq 0 \quad (x \in \mathbb{R}),$$

and

$$|e_i(x) - \delta_{i,j(x)}| \leq C h(x) \quad (x \in \mathbb{R},\ 1 \leq i \leq k). \tag{3}$$

Consequently, for every $n \geq 1$ and $x \in \mathbb{R}^n$, the quantities

$$s_i = \sum_{a=1}^{n} e_i(x_a), \qquad t = \sum_{a=1}^{n} h(x_a), \qquad m_i = \#\{a : j(x_a) = i\} \tag{4}$$

satisfy $m \in \mathbb{N}_0^k$, $t \geq 0$ and $|s_i - m_i| \leq Ct$. Every $m \in \mathbb{N}_0^k$ is realized with $s = m$ and $t = 0$ by a finite spectrum.

Proof. For $1 \leq i \leq k$, define

$$\ell_i(x) = \prod_{\substack{0 \leq j \leq k \\ j \neq i}} \frac{x - j}{i - j}, \qquad e_i(x) = [1 - 2\ell_i'(i)(x - i)]\,\ell_i(x)^2, \qquad h(x) = \prod_{j=0}^{k} (x - j)^2. \tag{5}$$

The squared Lagrange factors give $e_i(j) = \delta_{ij}$ and $e_i'(j) = 0$ at every $j \neq i$. At i, the two terms in the derivative cancel, so $e_i(i) = 1$ and $e_i'(i) = 0$ as well. The degree and zero-constant assertions follow from these formulas.

For every i, j the quotient

$$q_{ij}(x) = \frac{e_i(x) - \delta_{ij}}{(x - j)^2}$$

is a rational polynomial of degree at most $2k - 1$. Put $R = 2(k+1)$ and take

$$C = 1 + 2^{2k} \max_{i,j} \|q_{ij}\|_R + 2^{2k+2} \max_i (1 + \|e_i\|_1), \tag{6}$$

where $1 \leq i \leq k$ and $0 \leq j \leq k$. This is a computable rational number.

If $|x| \leq R$ and $j = j(x)$, every other node l satisfies $|x - l| \geq 1/2$. Indeed, a node at distance less than $1/2$ would be the unique closest node. It follows that

$$h(x) \geq 2^{-2k}(x - j)^2, \qquad |e_i(x) - \delta_{ij}| \leq (x - j)^2 \|q_{ij}\|_R \leq C h(x).$$

This also holds directly when $x = j$. If $|x| > R$, then $|x - l| \geq |x| - k > |x|/2$ for every node l. Therefore

$$h(x) \geq (|x|/2)^{2k+2}, \qquad |e_i(x) - \delta_{i,j(x)}| \leq (1 + \|e_i\|_1)|x|^{2k+1},$$

which again proves (3). Summing the pointwise bounds gives $|s_i - m_i| \leq Ct$ without a factor depending on n. Finally, take one zero entry and m_i copies of i for each $i = 1, \ldots, k$. This spectrum has size $1 + \sum_i m_i \geq 1$ and gives $s = m$, $t = 0$. □

3 The effective reduction

The following discrete decision problem is undecidable: given an integer polynomial, determine whether it is nonnegative on every tuple of positive integers [3, Section 5, p. 11 of the arXiv version]. This is a consequence of Matiyasevich's theorem used in that source. Replacing each variable by $u_i + 1$ shows that nonnegativity on $\mathbb{N}_0^k$, for arbitrary finite k, is also undecidable.

We first control the change in such a polynomial when the interpolated sums s are replaced by the associated integer counts m from (4).

Lemma 3. *Let $P(u) = \sum_\alpha c_\alpha u^\alpha \in \mathbb{Z}[u_1, \dots, u_k]$, where $|\alpha| = \sum_i \alpha_i$. Set*

$$D = \max\{1, \deg P\}, \qquad K = \sum_\alpha |c_\alpha|\,|\alpha|, \qquad L = 3^{D-1}C^2K^2,$$

with $D = 1$ when $P = 0$. If $t \geq 0$ and $|s_i - m_i| \leq Ct$ for every i, then

$$\bigl(P(s) - P(m)\bigr)^2 \leq Lt^2 \left(1 + \sum_{i=1}^{k} s_i^2 + C^2t^2\right)^{D-1}. \tag{7}$$

Proof. For a nonconstant monomial, telescope the difference of its $|\alpha|$ coordinate factors, including repetitions. Each changed factor contributes at most Ct, and the other factors have magnitude at most $\max_i |s_i| + Ct$. Thus

$$|s^\alpha - m^\alpha| \leq |\alpha| Ct \bigl(1 + \max_i |s_i| + Ct\bigr)^{|\alpha|-1}.$$

The base is at least 1. Multiplying by $|c_\alpha|$, summing, and using $|\alpha| \leq D$ gives

$$|P(s) - P(m)| \leq CKt \bigl(1 + \max_i |s_i| + Ct\bigr)^{D-1}.$$

The inequality $(1 + a + b)^2 \leq 3(1 + a^2 + b^2)$ and $(\max_i |s_i|)^2 \leq \sum_i s_i^2$ prove (7). For a constant polynomial, including $P = 0$, both K and the difference are zero, and the same bound holds with $D = 1$. □

Proof of Theorem 1. Given $P \in \mathbb{Z}[u_1, \dots, u_k]$, set $d = 2k + 2$ and take the polynomials and C of Lemma 2. Write

$$e_i(x) = \sum_{r=1}^{d} e_{ir} x^r, \qquad h(x) = \sum_{r=1}^{d} h_r x^r,$$

allowing zero coefficients. Define linear forms in the positive power-sum variables by

$$S_i(z) = \sum_{r=1}^{d} e_{ir} z_r, \qquad T(z) = \sum_{r=1}^{d} h_r z_r.$$

With D, K, L as in Lemma 3, form the rational polynomial

$$\begin{aligned} F_P(z) &= 2P\bigl(S_1(z), \dots, S_k(z)\bigr) + 1 \\ &\quad + LT(z)^2 \left(1 + \sum_{i=1}^{k} S_i(z)^2 + C^2 T(z)^2\right)^{D-1}. \end{aligned} \tag{8}$$

All these operations terminate in exact rational polynomial arithmetic: finite products, derivatives, exact divisions by $(x-j)^2$, rational coefficient bounds and finite compositions. The expression can be expanded into a finite coefficient list. No discrete solution is needed to construct it.

At $z = (p_1(x), \dots, p_d(x))$, the forms S_i and T equal the spectral sums s_i and t in (4). Since $e_i(0) = h(0) = 0$, these sums use only positive powers. The constants in (8) are scalar polynomial constants, so the construction never introduces $\operatorname{tr} I$ or a zeroth power-sum variable.

Suppose $P(m) \geq 0$ for every $m \in \mathbb{N}_0^k$. For any finite real spectrum, let m be its rounded counts and put $\eta = P(s) - P(m)$. Lemma 3 gives

$$\begin{aligned} F_P(p_1(x), \dots, p_d(x)) &\geq 2P(m) + 2\eta + 1 + \eta^2 \\ &= 2P(m) + (\eta + 1)^2 \geq 0. \end{aligned}$$

Conversely, if $P(m) < 0$ at some $m \in \mathbb{N}_0^k$, then $P(m) \leq -1$ by integrality. The node spectrum from Lemma 2 has $s = m$, $t = 0$ and therefore

$$F_P\big(p_1(x), \dots, p_d(x)\big) = 2P(m) + 1 \leq -1.$$

We have proved

$$P \geq 0 \text{ on } \mathbb{N}_0^k \quad \Longleftrightarrow \quad F_P\big(p_1(x), \dots, p_d(x)\big) \geq 0 \text{ for every } n \geq 1,\ x \in \mathbb{R}^n.$$

An always-terminating decider for (1), applied to the computable input $(2k+2, F_P)$, would decide the discrete problem. This contradiction proves the theorem. □

AI-use statement. OpenAI Codex was used to develop the argument, check cited sources and draft this manuscript. Independent human mathematical review is requested.

References

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